IoT-Based Poultry Farm Gas Monitoring System

Master's Research Project | Prairie View A&M University

Duration

June 2024 – May 2025

Technologies

Python • Flask • SQL • ThingSpeak • Raspberry Pi 4 • Machine Learning • Scikit-learn • HTML • CSS • JavaScript • REST API

Project Overview

Designed and developed a real-time IoT-based environmental monitoring system that continuously measures CO_2, CH_4, and NH_3 gas concentrations inside poultry farms. The solution combines IoT sensors, cloud computing, machine learning, and a web dashboard to help farm operators monitor air quality and predict hazardous gas levels before they occur.

Problem Statement

Traditional poultry farms rely on manual gas monitoring, which is time-consuming and cannot detect sudden increases in toxic gases. Poor air quality negatively impacts poultry health, productivity, and worker safety.

Solution

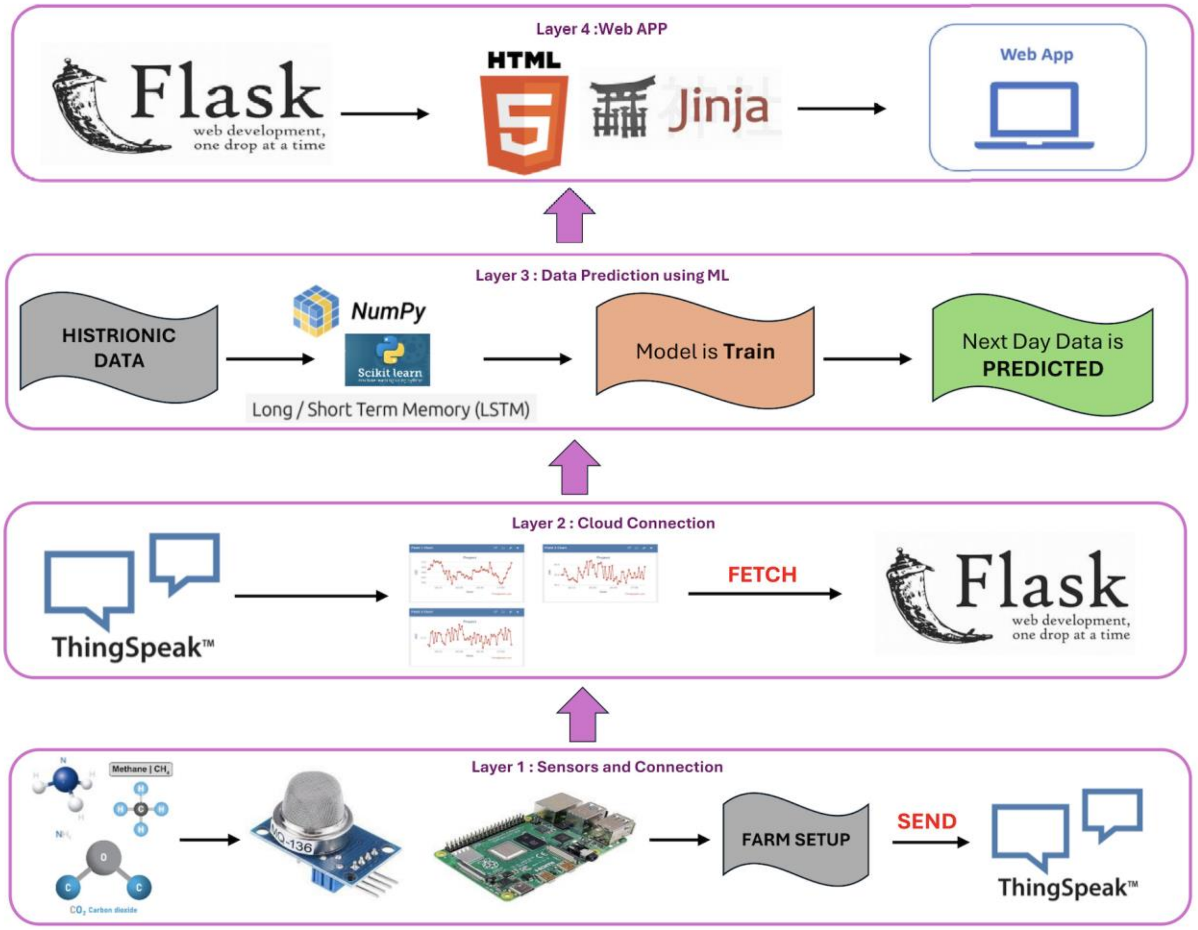
• Developed an end-to-end IoT monitoring platform using Raspberry Pi 4 and MQ-135 sensors
• Built REST API integration with ThingSpeak Cloud
• Created a Flask web application for real-time monitoring
• Implemented Machine Learning forecasting for next-day gas prediction
• Generated automatic alerts when gas thresholds were exceeded

Technologies

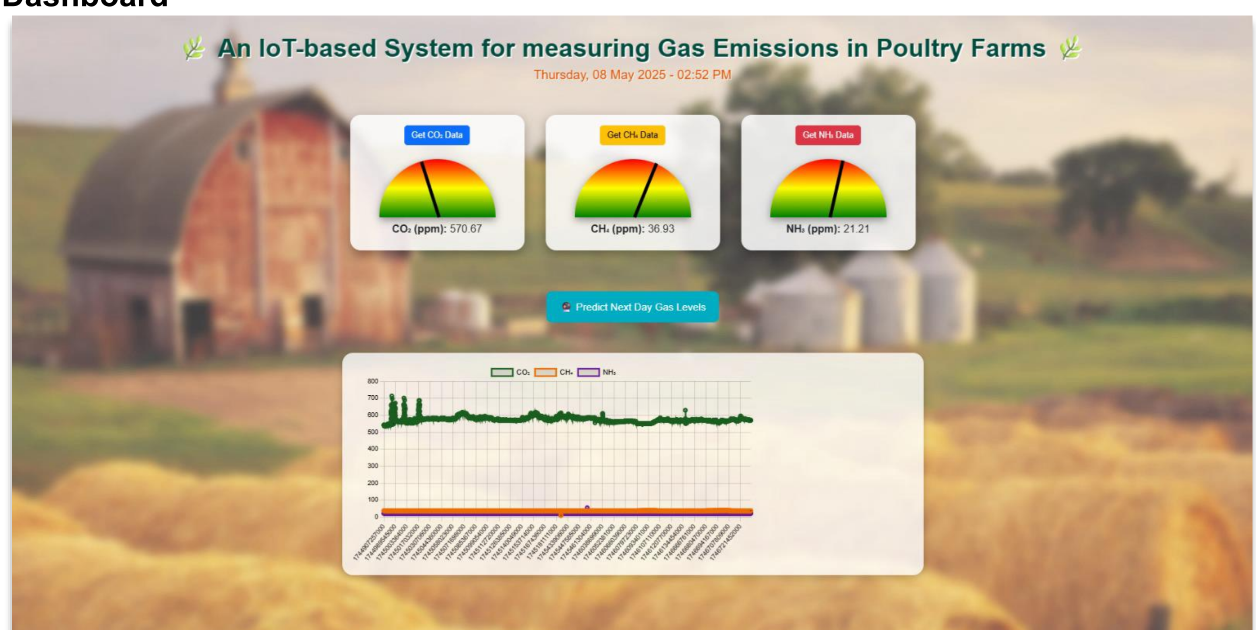
Python
Flask
SQL
ThingSpeak API
Raspberry Pi
Machine Learning
Random Forest
Pandas
NumPy
Scikit-learn
Plotly
HTML/CSS
REST API
Git

Architecture

Machine Learning



Dashboard



Machine Learning

Model: Random Forest Regressor
Prediction Horizon: Next 24 Hours
R^2 Score: **0.87**
MSE: **35.2**

Key Features

- Real-time monitoring
- Cloud synchronization
- Historical analysis
- Interactive dashboard
- CSV export
- Threshold alerts
- Machine learning prediction
- Responsive web application

Results

- Achieved **R^2 = 0.87** for gas prediction.
- Reduced manual monitoring through continuous real-time data collection.
- Built a scalable, low-cost solution suitable for smart agriculture deployments.
- Published the research as part of a **Master of Science in Computer Science** project.

My Contribution

- Designed the complete system architecture
- Developed the Flask web application
- Built the Python backend
- Developed the machine learning model
- Integrated Raspberry Pi sensors
- Connected ThingSpeak APIs
- Created interactive dashboards
- Tested and deployed the system

Research Paper: https://www.mdpi.com/2624-7402/7/8/267